

Challenge: Convert Data Types

Ensuring that each column in your dataset has the correct data type is fundamental for accurate analysis and computation. When columns meant to represent numbers are stored as strings, calculations and statistical operations may fail or produce incorrect results. This is especially common when importing data from CSV files, spreadsheets, or external sources, where values such as numbers or dates are sometimes interpreted as text. Converting these columns to their appropriate data types ensures that you can perform mathematical operations, aggregations, and visualizations without unexpected errors or misleading outcomes.

```
import pandas as pd

# Sample data where the 'price' and 'quantity' columns are stored as strings
data = {
    'item': ['apple', 'banana', 'orange'],
    'price': ['1.20', '0.80', '1.00'],
    'quantity': ['10', '25', '15']
}

df = pd.DataFrame(data)

print(df)
print(df.dtypes)
```